

Program 5: Demonstrate CI Integration by running Jenkins in a Docker container

Step 1: Create a directory and change to it

```
mkdir program5
```

```
cd program5
```

Step 2 : create dockerfile

```
nano Dockerfile
```

```
FROM jenkins/jenkins:lts

#switch to root user to install tools
USER root

#install docker inside jenkins container
RUN apt-get update && apt-get install -y docker.io

#switch back to jenkins user
USER jenkins
```

Step 3: Create docker-compose.yml

```
nano docker-compose.yml
```

```
version : '3.8'
services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

Step 4: build the container

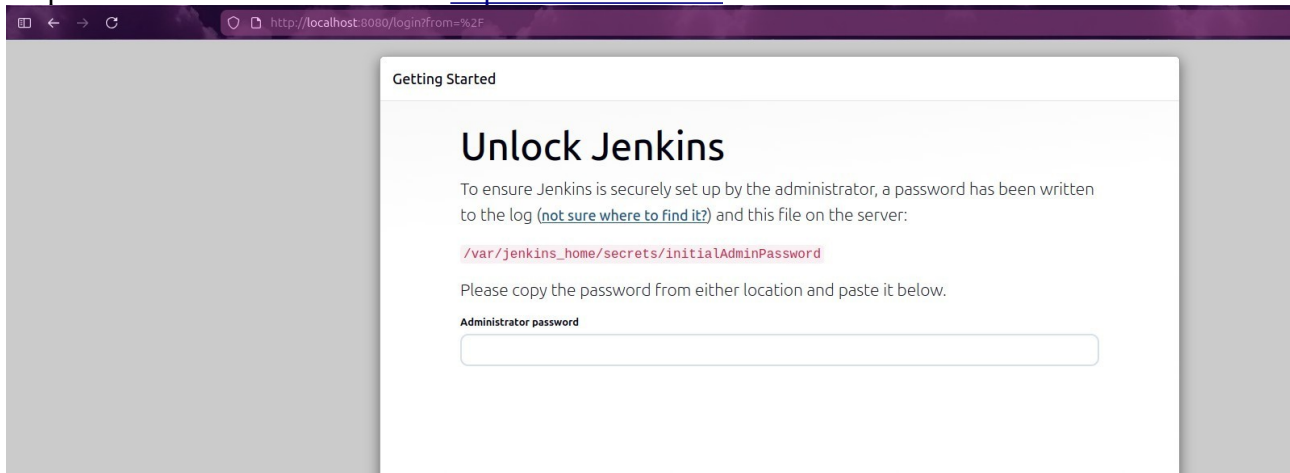
```
docker compose build
```

```
[+] Building 7.0s (9/9) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 539B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 249B 0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts 6.9s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:7b1c378278279c8688efd 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:f5df4847ab9add46eb89d97e1eb3b1a3b166db5b950bb 0.0s
=> => naming to docker.io/library/prgm5-jenkins 0.0s
=> resolving provenance for metadata file 0.0s
[+] Building 1/1
✔ prgm5-jenkins Built 0.0s
```

Step 5: Docker up

Step 6: List the images and check for the newly created image
docker images or docker ps

Step 7: Check in the browser with <https://localhost:8080>



Step 6: Remove the docker image of jenkins
docker image rm program5-jenkins:latest